Constraining Turbulence in the Intracluster Medium with the Radio Sunyaev Zeldovich Effect

by

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CURRENT DEGREES

Candidate's ORCID

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Te Herenga Waka - Victoria University of Wellington

2027

Abstract

Gas motions. SZ effect Pressure fluctuations. Stable methodology. modedling. simulation and some interferometry.

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Praise to Allah that has given me strength. Appreciation to my parents for their support. Thanks to my supervisors for their patience.

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For F.N.E. May she live forever.

Contents

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List of Abbreviations and Symbols

Abbreviations	
AC	Alternating Current
AFM	Atomic Force Microscopy/Microscope
etc.	*etc.*

Symbols	
$\hat{\rho}$	Density operator
etc.	*etc.*

Chapter 1

Scientific Landscape

> "If you wish to create a Universe from scratch, the final ingredient would be clusters of galaxies."
>
> — [Afiq Hamid 2025]

1.1 Galaxy Clusters

It has been said that mass and time are among the most fundamental parameters in all of astrophysical science. This thesis is principally an exploration of the former through the study of the most massive virialized objects known to exist in the Universe: galaxy clusters.

Galaxy clusters occupy a unique position within the hierarchy of cosmic structures. They represent the endpoint of cosmic structure formation, populating the nodes that connect the filaments of the cosmic web [ref]. The matter composition of galaxy clusters is dominated by a hitherto enigmatic, collisionless dark matter (DM) component ($\sim 80\%$), while the remaining baryonic constituents are made up of the intracluster medium (ICM), a diffuse and turbulent plasma ($\sim 15\%$), with the luminous member galaxies — themselves home to most of the stars, planets, and people within the cluster — make up the small residual ($\sim 5\%$) [ref].

Structurally the DM component dictates the overall potential well, setting the stage for all of the baryonic interactions [ref]. At the base of this immense potential well sits the largest galaxy known as the brightest cluster galaxy (BCG), that accounts for 0.5–2% of the cluster baryons. The BCG commonly hosts an active galactic nucleus (AGN), a powerful source of energetic feedback attributed to super massive black hole (SMBH) activity at the origin [ref]. Understanding AGN physics is of broad interest in modern astrophysics, spanning diverse research interests from exotic accretion processes to high powered jet emission [ref], [ref], and while not the primary focus of this work, AGN feedback has some implications for the stirring of gas motions within the ICM.

Permeating the space between the member galaxies in a cluster is the ICM. It is a hot ($T \sim 10^7$–10^8 K), diffuse ($n_e \sim 10^{-3}$–10^{-1} cm^{-3}), optically thin, volume-filling plasma. The ICM is threaded by a weak but dynamically relevant magnetic field ($\sim$ 1–10 μG), detectable via Faraday rotation measures [ref]. As the primary atmosphere of the cluster, the ICM plays an important role in various dynamical processes therein.

- ***AGN thermostat*** The ICM regulates the cooling of gas towards the cluster core — a process that, if left unchecked, would produce excess star formation rates (SFR) in the BCG. In the absence of a heating mechanism, the dense core would radiate energy efficiently and cool on timescales far shorter than the cluster age, driving a so-called cooling flow [ref]. This cooling is now understood to be moderated by a self-regulating feedback loop: as gas cools and accretes onto the BCG it fuels the central AGN, which injects energy back into the ICM via jets and lobes, suppressing further cooling of the core regions [ref]. The ICM therefore acts as both the fuel supply and calorimeter of this baryonic cycle.

- ***Star formation in member galaxies*** The ICM plays an influential role on the evolution and SFR of its member galaxies through quenching mechanisms and ram-pressure stripping (RPS). As the satellite galaxies fall into the cluster potential well, contact with the ICM removes internal cold gas reservoirs from the interstellar medium (ISM) of the galaxies, quenching star formation on relatively short timescales ($t \sim 0.3$ Gyr) [ref]. This is most evident on close approach to the core and manifests as the stripped tails of jellyfish galaxies [ref]. Conversely, on longer timescales ($t \sim 3$ Gyr) strangulation targets the outer hot gas halo and prevents new gas from cooling [ref], [ref].

- ***Cluster mass bias*** Cluster mass measurements can be used as cosmological probes of the large scale matter distribution of the Universe constraining several important parameters of the ΛCDM cosmology model [ref]. Since the DM component cannot be directly observed by any present means, the ICM provides the an observational avenue of constraining cluster masses via complementary multi wavelength approaches. However, the approach is heavily dependent on the assumption that the ICM gas density is in approximate hydrostatic equilibrium (HE) within the DM potential well [ref].

The observational leverage we have on all three of the aforementioned processes is mediated through how the ICM manifests across the electromagnetic spectrum. The following section details the primary observational windows through which the ICM is studied in order to derive constraints on the cluster mass, and introduces the key physical quantities and input assumptions that underpin the work done in later chapters.

1.2 Observables

Optical light as the oldest astronomical window was used to initially identify and catalog the first groups and clusters [ref]. It remains a viable avenue to evaluate the matter content of galaxy clusters via gravitational lensing [ref] and velocity dispersion of member galaxies [ref]. However the windows of most relevance towards the mapping of the internal thermodynamic states of the ICM are namely the X-ray and millimeter-wave radio respectively.

1.2.1 X ray

The X-ray window played a vital role in the initial discovery and characterisation of the ICM. To date the ICM of galaxy clusters remain among the brightest X-ray sources on the sky. The emission is predominantly produced via thermal bremsstrahlung (free-free emission), in which electrons are deflected by ions and radiate photons in the process. The X-ray surface brightness S_X of the ICM electrons is proportional to the line-of-sight (LOS) emission integral [ref],

$$S_\mathrm{X} \propto \int n_e n_p \Lambda(T)\,\mathrm{d}l \approx \int n_e^2 \Lambda(T)\,\mathrm{d}l, \tag{1.1}$$

where n_e and n_p are the electron and proton number densities respectively, $\Lambda(T)$ is the plasma cooling function, and the integral is taken along the line of sight z. Since the emissivity scales as n_e^2, X-ray observations are more sensitive to the dense cluster core, making them well suited for detailed high resolution studies of the inner ICM. However, this sensitivity can be a double-edged sword as the n_e^2 dependence means that S_X can be biased by an unknown gas clumping factor C_ρ arising from inhomogeneous gas distributions in the core [ref].

Since the sound crossing time (represented by the time it takes for pressure fluctuations to travel across a region in the ICM) within an Abell radius of 3 Mpc is only 0.1 - 0.3 of the Hubble time (the timescale of the cluster formation and evolution), it is generally assumed that the ICM will eventually settle into a state of approximate HE:

$$\frac{1}{\rho_g}\frac{dP}{dr} = -\frac{d\phi}{dr} = -\frac{GM(<r)}{r^2}, \tag{1.2}$$

where $M(<r)$ is the total mass enclosed within radius r, ϕ is the gravitational potential, ρ_g is the gas density, and P is Pressure. This local hydrostatic balance contributes to the cluster's broader virialized state and provides a foundational framework through which X-ray observations can be used to infer cluster mass. There has been overall good consensus between traditional X-ray methods in finding the true cluster mass with maximum root mean square (RMS) deviation ranging from 5 % to 20 % [ref] [ref] under the HE assumption . However, there remains certain inherent limitations of the X-ray approach that can be complemented by other wavelengths.

Another important concept that emerged from the X ray regime is the empirical β-model. By applying an approximation for a self-gravitating collisionless system (originally developed for globular clusters by [ref]) to a radial model of galaxy and gas distribution in equilibrium within the same potential [ref] derived the analytic isothermal radial β model of the ICM:

$$n_{gas}(r) = n_0 \left[1 + \left(\frac{r}{r_c}\right)^2\right]^{-\frac{3}{2}\beta}, \tag{1.3}$$

where r_c is the radius of the cluster core and $\beta \sim 2/3$ is known as the canonical β value [ref]. The profile describes a flat central core transitioning to a power-law decline at large radii. Its broad applicability across the cluster population follows naturally from the self-similar nature of galaxy clusters. While modern high-resolution observations and hydrodynamical simulations have revealed limitations of the model — particularly in its assumptions of isothermality and spherical symmetry, and its inability to capture the complexity of merging or dynamically disturbed systems — it remains a robust and widely used first approximation for the ICM radial gas density distribution [ref].

1.2.2 Radio Sunyaev-Zel'dovich effect

The thermal Sunyaev–Zel'dovich effect (tSZ) is a secondary CMB anisotropy arising from the inverse Compton scattering of CMB photons off energetic thermal electrons within the ICM [ref]. As CMB photons traverse the cluster, a small fraction are upscattered to higher energies, producing a characteristic spectral distortion: a decrement in CMB intensity at frequencies $\nu < 217$ GHz and a corresponding increment at $\nu > 217$ GHz. The resulting change in CMB intensity is given by [ref]:

$$\Delta I_\nu = y \frac{2h\nu^3}{c^2} \frac{xe^x}{(e^x - 1)^2} \left[x \frac{e^x + 1}{e^x - 1} - 4\right], \tag{1.4}$$

where $x = h\nu/k_B T$ is the dimensionless frequency, $T = 2.725$ K is the CMB temperature, h is Planck's constant, and k_B is Boltzmann's constant. The amplitude of the distortion is set by the Compton y parameter, which is proportional to the LOS integral of the electron pressure:

$$y = \frac{\sigma_T}{m_e c^2} \int P_e dl, \tag{1.5}$$

where $P_e = n_e k_B T$ and σ_T is the Thomson cross section. The Compton y parameter therefore provides a direct in-situ measure of the integrated thermal pressure of the cluster along the LOS. A key advantage of the tSZ signal over S_X is its linear dependence on electron density making it more sensitive to the low-density outskirts of the cluster towards the virial radius. Coupled with a sufficiently wide field of view (FOV), the tSZ effect is therefore particularly well suited to mapping the thermodynamic structure of the ICM on large scales and by extension cosmic baryons[ref]. Figure E.1 shows a multispectral image of the Coma cluster of galaxies from both tSZ and S_X.

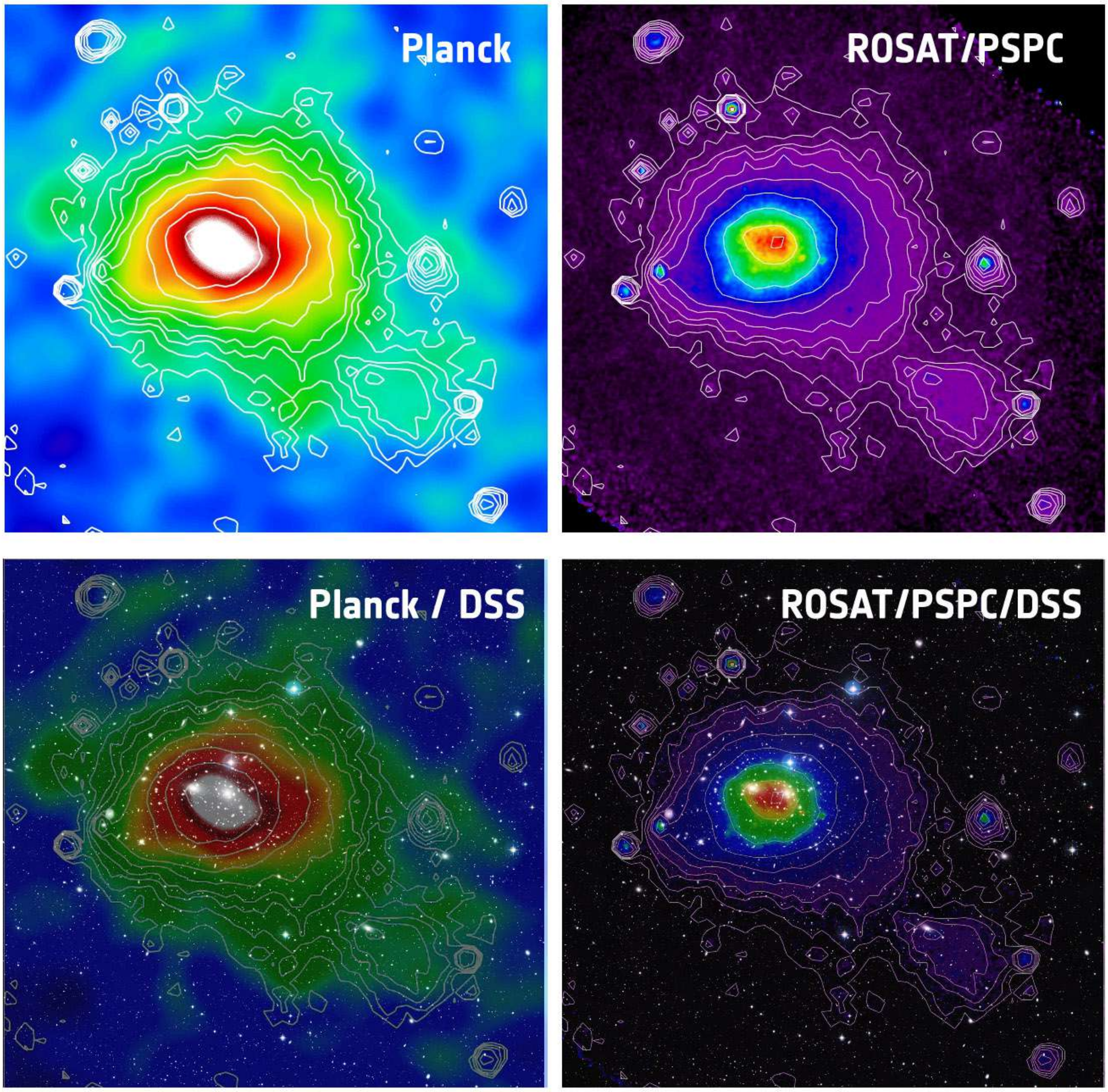


Figure 1.1: Multiwavelength composite of the Coma cluster (Abell 1656). Top left: Planck tSZ Compton y map. Top right: ROSAT soft X-ray surface brightness. In both upper panels the X-ray contours are superimposed on the tSZ image to illustrate the complementary spatial coverage of the two windows. Bottom panels show each overlaid on the Digitised Sky Survey optical image, revealing the member galaxy distribution. Image credit: ESA/LFI & HFI Consortia; Max-Planck-Institut für extraterrestrische Physik; NASA, ESA, and the Digitised Sky Survey 2 [ref] [ref].

1.3 ICM structure and non-thermal Pressure

Notwithstanding the overview provided thus far, the ICM is in reality a complex and dynamically evolving system. The true gas conditions are in actuality far removed from the idealised equilibrium states assumed in the foundational framework described in the previous sections. A truer conceptual picture — one favoured in the present literature — models the ICM as a stratified, multiphase fluid gravitationally confined within the cluster potential well [ref][ref]. We now briefly interrogate each of these concepts in turn.

A fluid description of the ICM

The treatment of the ICM as a fluid is suitable when the mean free path λ_{mfp} of the constituent particles is much smaller compared to the characteristic scale length L of the system. For the bulk ICM, $\lambda_{\mathrm{mfp}} \approx 4.6$ kpc [ref], which is well below the scales of interest for cluster-wide gas dynamics — justifying the use of the Euler or Navier–Stokes equations as a fluid description [ref].

However, this fluid approximation is not universal. In the low-density outskirts of clusters, where the mean free path is much higher, and on small scales where kinetic plasma effects become important, a purely hydrodynamic description breaks down and kinetic or magnetohydrodynamic (MHD) formulisms are more appropriate [ref].

Stratification

Stratification simply means there is a radial ordering of the different thermodynamic properties — principally density, temperature, and entropy all under the simultaneous influence of gravity [ref]. The condition of stratification is characterized by the Brunt–Väisälä (BV) frequency:

$$N_{\mathrm{BV}}^2 = \frac{g}{\gamma}\frac{d}{dr}\ln\frac{P}{\rho_{\mathrm{g}}^{\gamma}}, \tag{1.6}$$

where g is the local gravitational acceleration, γ is the adiabatic index, P is the thermal pressure, ρ is the gas density, and r is the radial coordinate. The quantity P/ρ^{γ} is proportional to the specific entropy K, thus N_{BV}^2 also describes an entropy gradient. When $N_{\mathrm{BV}}^2 > 0$ — as is generally the case throughout the bulk ICM where entropy increases outward — the stratification is stable: a radially displaced gas parcel experiences a restoring buoyancy force and oscillates about its equilibrium position at frequency N_{BV} rather than convecting freely. This suppresses radial mixing and supports the propagation of internal gravity waves (g-modes), which are themselves a source of turbulent energy [ref].

There are some edge cases that are beyond the purely hydrodynamic cases that can introduce exotic processes such as magneto thermal instabilities (MTI) which argue for conditions that could prove degenerative to these stable N^2 states[ref].

Multiphase structure

The ICM is also not characterised by a single thermal phase. Modern observations and simulations reveal a multiphase environment in which gas spans a wide range of temperatures — from cold molecular filaments at $T \lesssim 10^2$ K to the bulk hot phase at $T \sim 10^7$–10^8 K — occurring within the same gravitational potential [ref]. This stands in sharpg contrast to the traditional picture of a uniform hot atmosphere, and draws analogy to the multiphase structure of the ISM.

Cold phases and filaments in the ICM develop as a result of thermal instability and chaotic cold accretion (CCA) [ref]. The CCA process within cluster cores is analogous to precipitation: where the local cooling time drops below the free-fall time, gas is able to condense out of the hot phase and 'rain' onto the center as cold filaments and clouds [ref]. A key theoretical challenge of understanding this turbulent process is trying to figure out how the cold filaments are not evaoprated back into the hot phase [ref]. The current candidates are a mix of anisotropic thermal conduction, unique magnetic field alignments, and heat flux buoyancy instabilities (HBI).

1.3.1 Non -thermal pressure

Within the stratified, multiphase fluid described of the ICM a range of dynamical phenomena consistently occur that introduce an additional layer of complexity to the base HE assumption. These challenges come in the form of bulk motion, turbulence, cosmic rays, and magnetic fields that all contribute additional non-thermal pressure components. In reality, the total pressure support within the ICM is not provided by thermal component alone:

$$P_{\rm tot} = P_{\rm th} + \frac{1}{3}\rho_{\rm g}\sigma_v^2, \tag{1.7}$$

where σ_v is the velocity dispersion of isotropically moving gas decomposable into radial and tangential components [ref]. Turbulent gas motions represent the dominant non-thermal contribution, with cosmic rays and magnetic fields contributing only at the level of a few per cent [ref]. From comparisons between theory and observation, residual gas motions are estimated to contribute between $\sim 10\%$ at R_{500} and $\sim 30\%$ at R_{200} of the total pressure support, with this fraction increasing in dynamically disturbed systems [ref].

Constraining $P_{\rm NT}$ observationally is a non-trivial problem. Direct constraints are in principle accessible through X-ray spectroscopy, with velocity broadening and shifts of spectral emission line methods [ref][ref]. However, the spatial coverage of current X-ray spectroscopic instruments is limited, and the technique is restricted to relatively nearby, X-ray bright systems as a result of redshift dependence. The SZ surface brightness fluctuation methods investigated in this work offer an indirect but spatially extended alternative, at the cost of a greater number of modeling assumptions — the characterization and interrogation of which are attempted throughout the next two chapters.

1.3.2 Astrophysical sources of gas motions.

The non-thermal pressure within clusters is sustained via continuous injection of kinetic energy into the ICM. The injection originates from a hierarchy of physical drivers operating across a wide range of spatial scales [ref]. These can be broadly ordered from large to small as follows.

On the largest scales, the dominant energy source is gravitational — the hierarchical assembly of large-scale structure through major cluster mergers and the continuous accretion of unvirialized material along cosmic web filaments [ref]. Major mergers between clusters of comparable mass drive powerful shocks and large-scale bulk flows that inject kinetic energy on scales approaching the virial radius, with the resulting turbulence taking of order a Gyr to equilibriate. Evidence from simulations shows the $P_{\rm th}$ to P_{NT} fraction rising up to 30% in the outskirts near R_{200} [ref]. Figure 1.2 presents a gallery of merging clusters to illustrate the phenomenon.

On intermediate scales, minor mergers and glancing interactions on the scale of galaxy groups drive sloshing motions that set the cluster atmosphere into coherent oscillations without fully disrupting the cool core [ref]. These produce cold fronts — sharp contact discontinuities between the displaced cool core gas and the surrounding hotter ICM — visible as edges in $S_{\rm X}$ [ref]. The sloshing flows are indicated to be subsonic corresponding to velocities of several hunder kilometres per second and mach numbers in the range of 0.3-0.5 [ref]. The infall of dark matter subhaloes further stirs the ICM on scales intermediate between the merger and core scales, contributing gaseous wakes from RPS and turbulent substructure [ref].

On the smallest scales, AGN feedback injects mechanical energy directly into the cluster core via jets and radio-mode outflows, inflating cavities visible in $S_{\rm X}$ and driving weak shocks that propagate outward into the surrounding ICM [ref]. The buoyantly rising cavities transfer most of their energy to the gas after crossing several pressure scale heights of the atmosphere [ref]. Additional contributions arise from plasma instabilities — including MTI and HBI — that operate in the weakly magnetised ICM and can drive small-scale anisotropic turbulent motions, particularly in the cluster outskirts [ref].

The cumulative effect of this multi-scale energy injection is a turbulent velocity field and fluctuations that permeates the ICM from the virial radius inward. In the next section we consider discus some fundamental physical aspects of the turbulence phenomena.

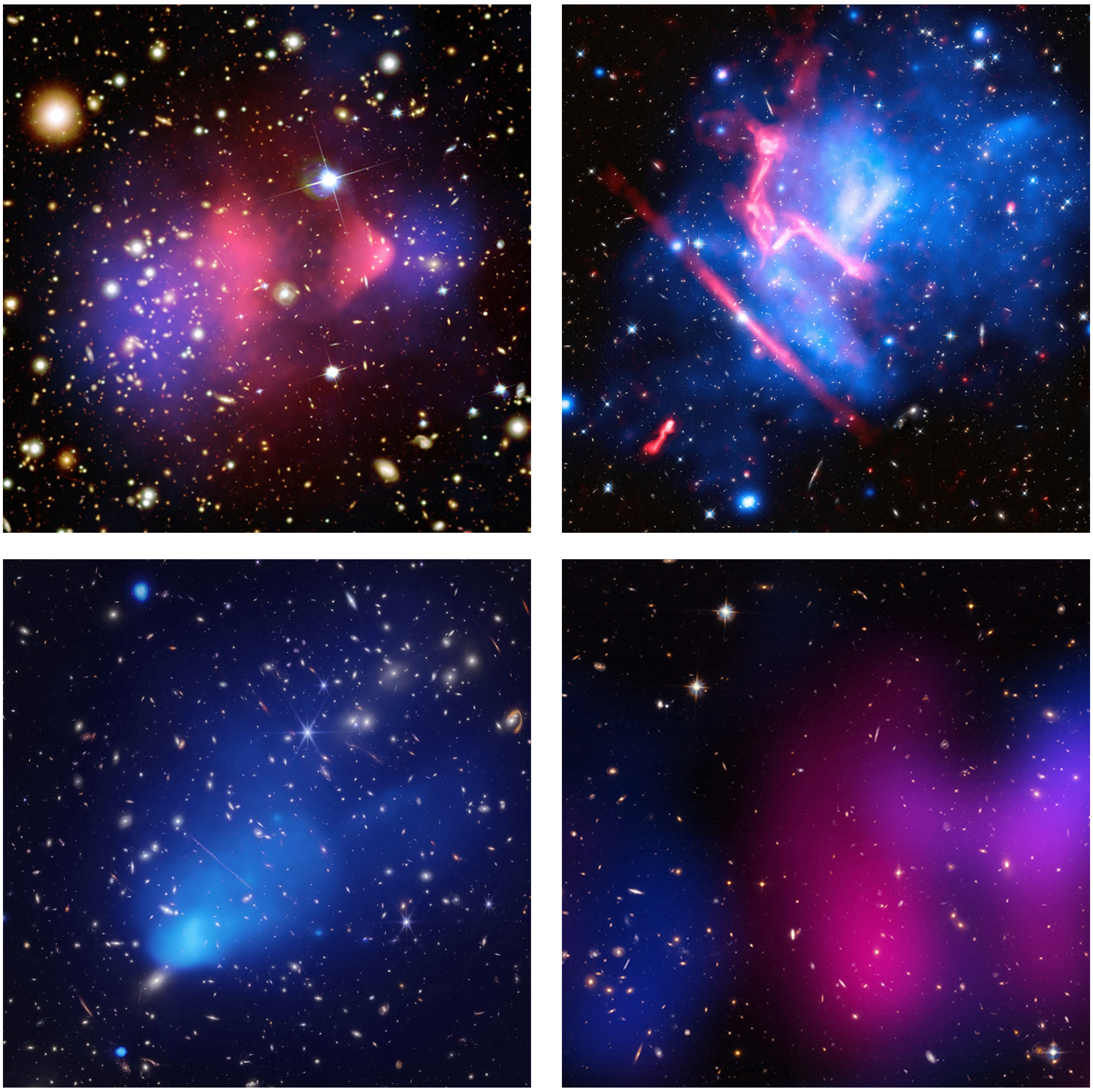


Figure 1.2: A gallery of dynamically disturbed galaxy clusters representing extreme violations of hydrostatic equilibrium. *Top left:* The Bullet Cluster (1E 0657-56), the archetypal binary merger. Chandra X-ray emission (pink) traces the shocked ICM stripped from the collisionless dark matter (blue, lensing reconstruction), providing the first direct observational evidence for dark matter.[ref]. *Top right:* MACS J0717.5+3745, the result of a quadruple cluster collision fed by a 13 Mpc cosmic web filament.Chandra X-ray (blue), VLA radio (red), and HST optical (background) reveal an extraordinarily complex thermodynamic structure [ref]. *Bottom left:* El Gordo (ACT-CL J0102-4915), the most massive known merging cluster at $z \sim 0.87$. Chandra X-ray emission (blue) is overlaid on a JWST infrared composite, with gravitational lensing arcs visible in the background field [ref]. *Bottom right:* The Musket Ball Cluster (DLSCL J0916.2+2951), a post-merger system caught further along in its evolution than the Bullet Cluster. The clear spatial offset between the hot gas (red) and dark matter (blue) illustrates the different collisional cross-sections of baryonic and dark matter [ref].

1.4 A short section on Turbulence theory

Turbulence is the study of irregular, chaotic fluid motion — one of the oldest unsolved problems in classical physics because of the non linear nature of its fundamental equations. In the terrestrial context it governs everything from the drag on aircraft wings to the mixing of cream in coffee. In the astrophysical context, the problem becomes even richer and more challenging: the flows are highly compressible, the scales are vast, boundary conditions are largely absent, the fluid is self-gravitating, and the presence of magnetic fields introduces additional complexity via magnetohydrodynamics (MHD). The ICM represents one of the most extreme turbulent environments in the Universe, and the true nature of its turbulent velocity field remains a big mystery.

The fundamental condition that allows for turbulence is set by the Reynolds number:

$$\mathrm{Re} = \frac{uL}{\nu}, \tag{1.8}$$

where u is the characteristic velocity of the flow, L is the driving scale, and ν is the kinematic viscosity of the fluid. Viscosity can be thought of as the amount of restiveness of the fluid. When $\mathrm{Re} \gg 1$, inertial forces dominate and viscous dissipation is negligible on the driving scale — the necessary condition for a turbulent flow to develop. For the ICM, estimates of the hydrodynamic Reynolds number is $\mathrm{Re} \leq 100 - 10^3$ and the magnetic Reynolds number is high $\mathrm{Re_m} \sim 100 - 10^3$ [ref]. Thus if you looked at the ICM on a purely fluid basis you would see a smooth slow pouring almost laminar fluid while looking at the magnetic fields reveals a chaotic haywire configuration.

1.4.1 The Kolmogorov energy cascade

The energy cascade is one of the core principles of turbulence theory. It describes a model of turbulence where kinetic energy is injected into the fluid at a large driving scale l_{inj} — in the ICM, this corresponds to the scale of the dominant energy source, whether a major merger, sloshing motion, or AGN outflow. However, the energy does not dissipate immediately, it is instead transferred progressively to smaller and smaller scales through the nonlinear interaction and fragmentation of eddies, with each generation of eddies feeding the next, until the viscous dissipation scale l_{diss} where the local Reynolds number approaches unity and viscous forces convert the remaining kinetic energy irreversibly into heat. The range of scales between l_{inj} and l_{diss} constitutes the *inertial range*, within which the process of energy transfer occurs. For astrophysical scales this range often spans many orders of magnitude [ref]. The power spectrum of this energy cascade is visualized in Figure 1.3.

Of the three scales of the energy cascade, the inertial range is of particular importance. It is within this range that the statistical properties of different turbulence models emerge. For an eddy of size l and characteristic velocity u_l the time required for fluid motions to traverse the eddy is:

$$t_{\mathrm{eddy}}(l) = \frac{l}{u_l}, \tag{1.9}$$

In the inertial range, where energy transfer is local in scale and the energy flux $\varepsilon \sim u_l^2/t_{\rm eddy}(l)$ is constant across scales, this gives the Kolmogorov velocity scaling $u_l \propto l^{1/3}$. The corresponding three-dimensional power spectrum of kinetic energy takes the form:

$$E(k) \propto C_p \dot{\varepsilon}^{2/3} k^{-5/3}, \tag{1.10}$$

where C_p is the universal non-dimensional Kolmogorov constant and $\dot{\varepsilon}$ is the turbulent energy dissipation rate per unit mass. This is celebrated but this is also angle averaged also known as the omnidirectional spectrum.

Yes you can get to Kolmogorov. But you need to talk a little more here. Its about 4-5 steps to Kolmogorov but you did it in only 2. Take your time and explain. So that ICM context can be added. How long it will take us to learn this derivation?

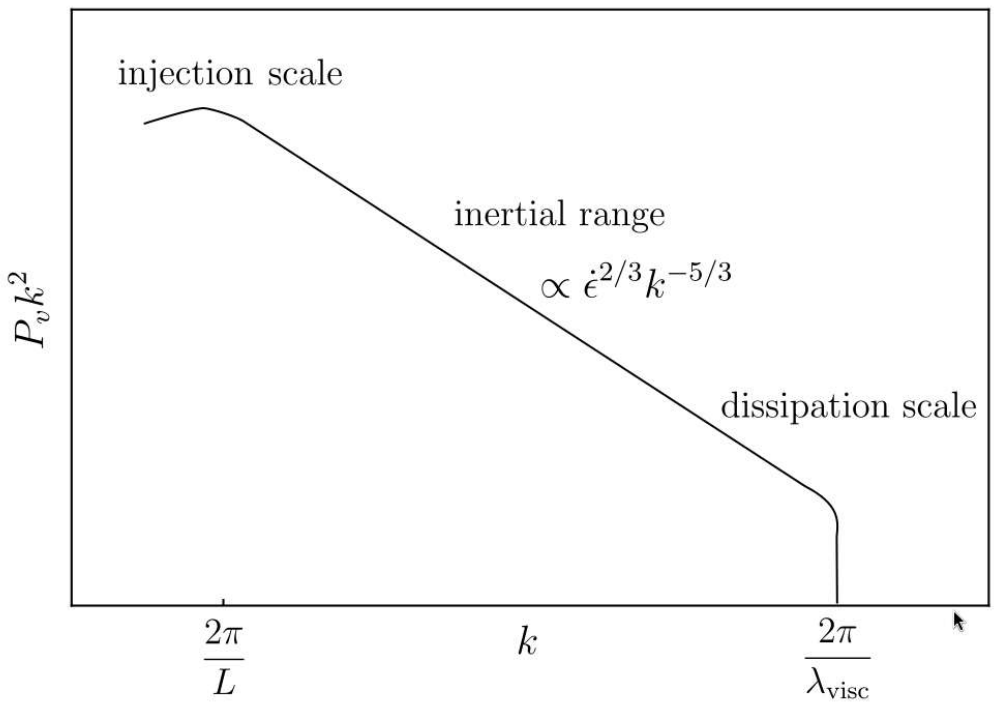


Figure 1.3: The three-dimensional velocity power spectrum $P_v k^2$ as a function of wavenumber k, illustrating the Kolmogorov turbulent energy cascade. Energy is injected at the driving scale $2\pi/L$, transferred through the inertial range with the characteristic $\propto \dot{\varepsilon}^{2/3} k^{-5/3}$ scaling, and dissipated at the viscous scale $2\pi/\lambda_{\rm visc}$. In the ICM context, the injection scale corresponds to the dominant energy driver — merger shocks, sloshing, or AGN activity — while the dissipation scale is set by the effective viscosity of the weakly collisional plasma.

Be careful that the 5/3 is angle averaged. Versus the 11/3 is the raw 3d modal. Why we take angle averaged because it is close to what we see in measurement.

So what the fuck are we writing here? Some more shit about turbulence? Right so sounds straight forward right? Use a telescope or spectroscopic measurement device, Observe this shit in space and constrain all the turbulence? But as people are prone to say yeah na. It is not so straightforward.

1.5 Mass Calibration and bias from non thermal pressure

So we talked alot of game on ICM physics and non thermal pressure as deviations for HE. Now we talk about their implications on mass measurements.

1.6 Objectives moving forward

Our science Objectives.

The following publication has been incorporated as Chapter **??**.

1. [**?**] **A. Hamid**, Co-author 1, and Final Author, Non parametric model and simulation of SZ surface brightness, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 2

Assumptions and Systematics in ICM Turbulence Inference from Surface Brightness Fluctuations

> "Real galaxy clusters exhibit a troubling awareness of our assumptions."

The inference of turbulent pressure support from surface brightness observations is not a direct measurement — it is the end product of a chain of modelling steps, each of which introduces assumptions about the geometry, thermodynamic state, and statistical properties of the ICM. This chapter provides a structured examination of the key assumptions and inputs that constitute the turbulence inference pipeline from surface brightness data.

2.1 Smooth SB model

The first step in the turbulence inference pipeline is the construction of a smooth model $\bar{S}(\theta)$ of the cluster surface brightness. The conceptual foundation for this decomposition traces back to the Reynolds decomposition of turbulent fluid mechanics, in which any instantaneous flow quantity is separated into a mean component and a fluctuating perturbation about that mean. Applied to the observed surface brightness $S_{\rm obs}(\theta)$, this decomposition takes the form:

$$S_{\rm obs}(\theta) = \bar{S}(\theta) + \delta S(\theta)$$

Where $S_{\rm obs}(\theta)$ is the observed surface brightness, $\bar{S}(\theta)$ is the smooth SB model, and $\delta S(\theta)$ is a small fluctuating component. The physical interpretation is that $\bar{S}(\theta)$ encodes the large-scale, slowly varying emission of the ICM in approximate hydrostatic equilibrium, while $\delta S(\theta)$ captures perturbations about that equilibrium — arising from turbulent density and pressure fluctuations,

This technique was first applied on Coma cluster in X ray. Any structure in the cluster that is not captured by $\bar{S}(\theta)$ will be absorbed into $\delta S(\theta)$ and misidentified as a turbulent fluctuation. The question is, is it a splotch (β model), a pancake (averaged empirical sphere), an egg (averaged empirical ellipse), a series of nested eggs (isocontour fit), or strained spagetti (Low Pass Filtered image)

2.1.1 Spherical Symmetry

In actuality this phenomena is nothing new. You are not the first to probe it. And by probe I mean quantify the amount of error that arises from this assumption. It has been known to have contributed error a small amount according to Pratt the error budget for this is roughly 1 percent. Haram you write a paper and thesis about cluster without citing the Buote papers. Read those papers, rabbit hole a bit, and cite here.

Galaxy clusters are not spherical, but almost everyone treats them as if they are, because it's computationally far simpler. The question is — how much does this matter, and for what observables?

Deviations from spherical symmetry, such as intrinsic flattening and substructure, will necessarily introduce scatter, and possibly biases, into spherically averaged global scaling relations used in cosmological studies.

This is the shit that you gotta curve fit to: This is the zeroth order model. This creates the flat beta. Extension to the zeroth order model is spherical and elliptical averaging. That is how you get empirical.

$$S_X(b) = S_0 \left[1 + \left(\frac{b}{r_c}\right)^2\right]^{-3\beta+0.5} \tag{2.1}$$

But how we do this is the

2.2 Turbulent Parameters

The ratio of density to Pressure is 1 which is pretty remarkable. However this is conditional on the thermodynamic state.

So once we have the raw fluctuations. we would then take the power spectrum. Then we deproject the power spectrum. We do not really know the true parameters. Suffice to say that shit is complex.

You have multi scale, multi layer, multi mach numbers. How does one even know what the injection scale rea;;y is? Here is some theory at least one page to prove it.

Kolmogorov turbulence [ref] applies in the subsonic, weakly compressible limit ($\mathcal{M} \ll 1$) where density fluctuations are small and the $E(k) \propto k^{-5/3}$ spectrum holds. For the bulk ICM with $\mathcal{M}_{3D} \lesssim 0.5$ this is a reasonable zeroth-order approximation [ref].

Burgers turbulence [ref] applies in the supersonic, shock-dominated limit ($\mathcal{M} \gg 1$) where the flow is dominated by discontinuous shocks rather than smooth eddies. The energy spectrum steepens to $E(k) \propto k^{-2}$. While the bulk ICM is subsonic, merger-driven shocks can locally reach this regime, particularly at cluster outskirts and along merger axes.

Iroshnikov–Kraichnan (IK) turbulence [ref] generalises the Kolmogorov framework to isotropic MHD. The presence of a mean magnetic field introduces Alfvén wave propagation as an additional timescale competing with the eddy turnover time, modifying the energy spectrum to $E(k) \propto k^{-3/2}$. Given that the ICM is magnetised at the level of ~ 1–$10\,\mu$G, MHD effects cannot be neglected entirely.

Goldreich–Sridhar (GS) turbulence [ref] extends the MHD description to the anisotropic case, where turbulent eddies are elongated along the local magnetic field direction. In this framework the cascade is critically balanced between the eddy turnover time and the Alfvén wave period, and the energy spectrum along and perpendicular to the field differs. This anisotropy is particularly relevant for the interpretation of ICM turbulence in the presence of stratification and ordered magnetic field geometries.

PITSZI:

$$P_{3D}(k) = \sigma_P^2 \frac{k_{3D}^2 \exp\left(-\frac{1}{k_{3D}^2 L_{inj}^2}\right) \exp\left(-k_{3D}^2 L_{diss}^2\right)}{\int 4\pi k_{3D}^{\alpha+2} \exp\left(-\frac{1}{k_{3D}^2 L_{inj}^2}\right) \exp\left(-k_{3D}^2 L_{diss}^2\right) dk_{3D}} \tag{2.2}$$

$$P_{3D}(k) = \sigma_P^2 \frac{k_{3D}^2 \exp\left(-\frac{1}{k_{3D}^2 L_{inj}^2}\right) \exp\left(-k_{3D}^2 L_{diss}^2\right)}{\int 4\pi k_{3D}^{\alpha+2} \exp\left(-\frac{1}{k_{3D}^2 L_{inj}^2}\right) \exp\left(-k_{3D}^2 L_{diss}^2\right) dk_{3D}} \tag{2.3}$$

2.3 Deprojection Process

The ratio of density to Pressure is 1 which is pretty remarkable. However this is conditional.

Everything we see is a two dimensional deprojection of a three dimensional phenomena. This invariable leads to some of information. Projection is simply an integral.

There is an Abel transfrom used for

2.4 Masking and Apodization

Masking is the filling of image pixels with a certain value. We do this in cluster science to isolate the ontarget shape away from the background. It is also a tendency to mask the core because the core is centrally condensed.

Apodization is the smooth tapering of the edges to prevent wringing. Roll off too quickly and you look like a hard edge. The question is what roll off is best roll off.

2.5 Instrumental Beam Effects

Whenever the sky is observed with a radio telescope some amount of blur is applied to the image. This is analogous to an effect of convolution with a 2D Gaussian.

Hello I am here. The following publication has been incorporated as Chapter 3.

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 3

Methodologies for Testing Turbulence Recoverability

In this chapter we talk about how we interrogated all the assumptions that go into the pipeline.

3.1 Non Parametric Methods

Without any prior assumptions on turbulent mach numbers.

3.1.1 Iso Contour Surface Brightness

Its a type of harmonics based algorithm. It interpolates concentric ellipses.

3.1.2 Fourier Mode Smooth models

This is actually a Low Pass Filtering

3.1.3 Ellipse models

This is actually two contributions in one. One is a sniper genetic algorithm method based method. And the other is a drone swarm.

Genetic Algorithms with Differential Evolution

Bayesian Inference with emcee

Probably some corner plots here. Talk about the Likelihood function.

3.2 fBm Fields and Artificial Clusters

Bayesian Inference with emcee

Beam Instrumentation

Hello I am here. The following publication has been incorporated as Chapter 4.

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 4

Methodologies for Testing Turbulence Recover ability

In this chapter we talk about the assumptions.

4.1 Introduction

Add your text here.

Chapter 5

Conclusion

Conclude your thesis.

Appendix A

Appendix

Write your appendix here. Following two are examples.

A.1 Name of Appendix-1

A.2 Name of Appendix-2

Appendix B

Example of Citations

This text is only for Bibliography testing purposes. This is a book by Cawvey et al. [**?**]. There are few journal articles [**?**, **?**] in this bibliography. This is a proceeding paper [**?**]. This is a PhD thesis by Peerling [**?**]. These are few other types of citations [**?**].

Appendix C

Example of Code

C.1 Find the greatest number from a list of numbers in *Python*

```
a=[1,2,3,4,6,7,99,88,999]
    max= 0
    for i in a:
        if i > max:
            max=i
    print(max)
```

Appendix D

Example of Equations

$$E = mc^2 \tag{D.1}$$

$$a = b + c \tag{D.2}$$

$$x = y + z \tag{D.3}$$

$$\begin{aligned} a = {} & b + c \\ & + d + e \end{aligned}$$

$$sinx + cosx = 1 \tag{D.4}$$

Appendix E

Example of Figures

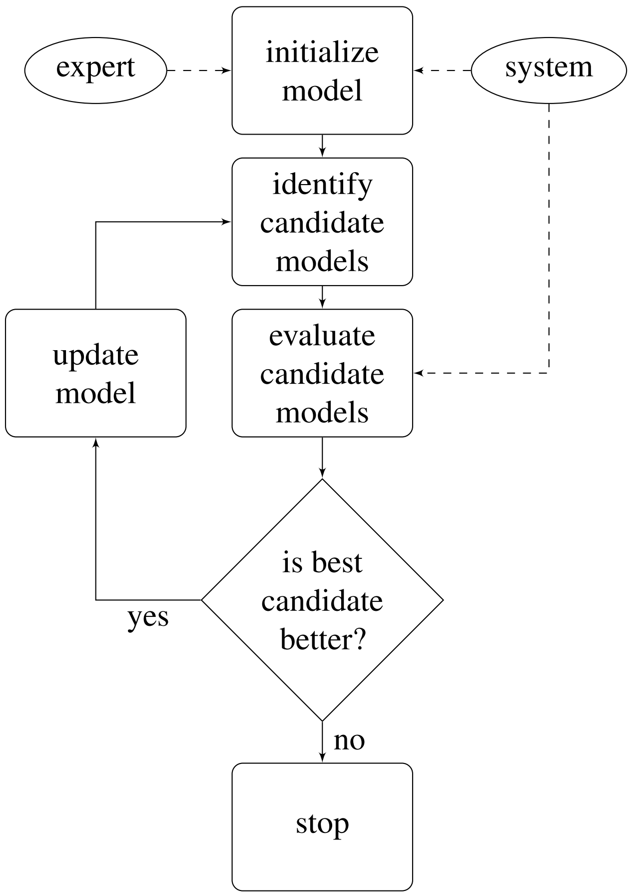
Figure E.1: The University Of Queensland

The Figure E.1 represents beauty of the UQ campus.

Appendix F

Example of Flow Charts

Figure F.1: Flow Chart



Flow chart F.1 is a simple example.

Appendix G

Example of Tables

Here is a really simple table G.1.

Table G.1: Name of the Australian Cities

Number	**Name**
1	Brisbane
2	Sydney
3	Melbourne
4	Canberra
5	Perth
6	Adelaide
7	Hobart
8	Darwin

Endquote goes here.

Author of quote,
Source of quote